Online Resource 2

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Article title: Progression from myeloneuropathy to cerebral disease in men with X-linked adrenoleukodystrophy studied using a Human Phenotype Ontology-based approach

Authors: Senol Dogan, Wolfgang Köhler, Caroline Bergner, Julia Lier, Sina Sadeghi, Toralf Kirsten

Corresponding author: Senol Dogan, Institute for Medical Informatics, Statistics and Epidemiology, Medical Faculty, Leipzig University, Härtelstr. 16-18, 04107 Leipzig, Germany. E-mail: Senol.Dogan@medizin.uni-leipzig.de

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Supplementary Methods

Handling of negative entries and visit dates

Many examination fields of the registry store free text. In the seizure field, 344 of 358 visit entries stated that the patient had no epilepsy and no seizures. A simple rule that counts every non-empty entry as a finding would class 127 of 215 patients (59.1%) as having seizures. With the rule applied in this study, only 2 patients had a seizure finding, and the other four cerebral subgroups did not change (Table S1 and Fig. S1). The effect was seen in every baseline profile (Table S2). Visit dates also matter. If undated visits are kept and sorted last, they can enter the first visits of a patient. In 52 of the 154 men this would set the follow-up time to 0.1 months. For this reason only dated visits were used in all analyses.

Supplementary tables and figure

Table S1 Number of patients with each cerebral HPO subgroup under a simple rule and under the rule applied in this study

HPO subgroup	All patients, simple rule	All patients, applied rule	Men, simple rule	Men, applied rule
Cognitive	38	38	35	35
Psychiatric	32	32	23	23
Seizure	127	2	89	2
Cranial nerve	22	22	17	17
Movement or ataxia	54	54	43	43

The simple rule counts every non-empty entry as a finding. The applied rule counts explicit negative entries as absent. Counts cover the first five visits of each of the 215 patients with a myelopathic start, of whom 154 were men.

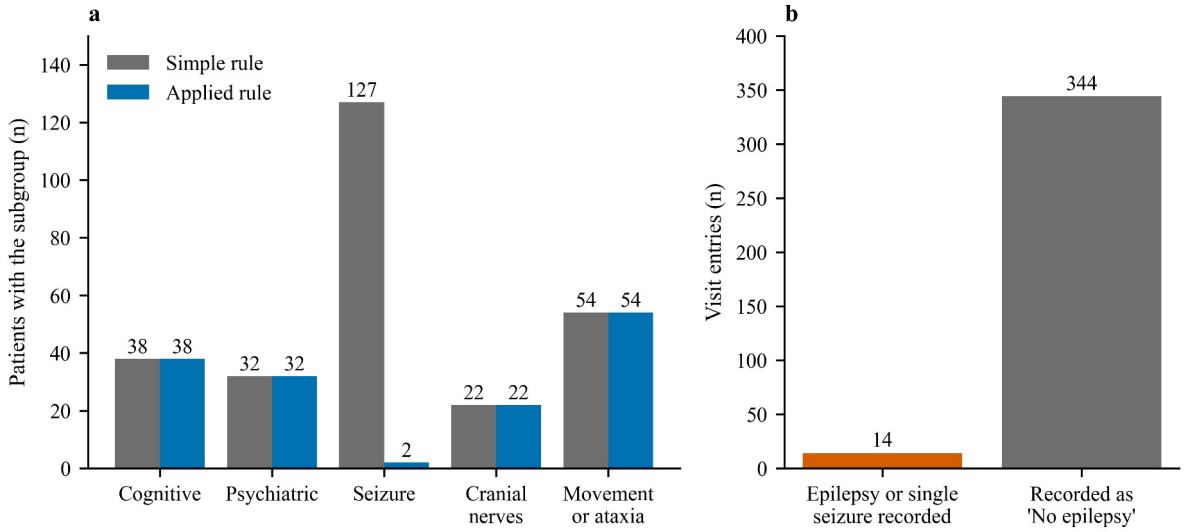


Fig. S1 Effect of the handling of negative entries. a Number of patients counted in each cerebral HPO subgroup under the simple rule and under the applied rule (n = 215). b Entries of the seizure field in all visit forms, split by their meaning

Table S2 Patients with a seizure finding by baseline myelopathic profile under the simple rule and the applied rule

Baseline profile	Patients, n	Simple rule, n (%)	Applied rule, n (%)
Autonomic	16	6 (37.5%)	0 (0.0%)
Autonomic + Motor/Gait/Balance	24	14 (58.3%)	0 (0.0%)
Autonomic + Motor/Gait/Balance + Sensory	6	5 (83.3%)	0 (0.0%)
Motor/Gait/Balance	136	81 (59.6%)	2 (1.5%)
Motor/Gait/Balance + Sensory	18	11 (61.1%)	0 (0.0%)
Sensory	15	10 (66.7%)	0 (0.0%)

Table S3 Entries of the seizure field in the visit forms of the 215 patients

Entry	Meaning under the applied rule	Visit entries, n
No epilepsy, no epileptic seizures	Counted as absent	344
History of singular epileptic seizure(s), no epilepsy	Counted as present	8
Diagnosis of epilepsy	Counted as present	6

Table S4 Penalised multivariable Cox models (ridge penalty 0.1)

Model	Factor	Hazard ratio (95% CI)	p
Profile model (ridge 0.1)	Sensory symptoms at start	0.82 (0.31 to 2.18)	0.69
Profile model (ridge 0.1)	Motor, gait or balance symptoms at start	0.24 (0.05 to 1.22)	0.086
Profile model (ridge 0.1)	Autonomic symptoms at start	0.47 (0.18 to 1.20)	0.11
Clinical model (ridge 0.1)	Age at first visit (per 10 years)	0.91 (0.62 to 1.35)	0.64
Clinical model (ridge 0.1)	Adrenal insufficiency documented	1.33 (0.60 to 2.94)	0.48

Table S5 Discrimination of the Cox models measured by the concordance index (C-index)

Model	Apparent C-index	Optimism	Corrected C-index	Bootstrap samples
Profile model	0.67	0.054	0.616	300
Clinical model	0.577	0.048	0.529	300
Age only	0.543	0.02	0.523	300

A C-index of 0.50 means no discrimination.

Table S6 Smallest hazard ratio detectable with 80% power

Factor	Prevalence	Events	Smallest detectable hazard ratio
Sensory symptoms at start	0.164	20	5.43
Motor, gait or balance symptoms at start	0.91	20	8.97
Autonomic symptoms at start	0.269	20	4.11
Adrenal insufficiency documented	0.537	20	3.51

Table S7 Sensitivity analyses and log-rank tests

Analysis	Men	Events	Sensory profile, HR (95% CI)	Autonomic profile, HR (95% CI)	Age per 10 years, HR (95% CI)
Primary analysis with all follow-up and any CALD record	67	20	0.86 (0.28 to 2.63)	0.36 (0.10 to 1.25)	0.83 (0.51 to 1.37)
Without records of arrest after transplantation only	67	20	0.86 (0.28 to 2.63)	0.36 (0.10 to 1.25)	0.83 (0.51 to 1.37)
First five visits only	67	11	0.94 (0.20 to 4.48)	1.11 (0.29 to 4.24)	0.66 (0.32 to 1.35)
At least 12 months of follow-up	66	20	0.86 (0.28 to 2.63)	0.36 (0.10 to 1.25)	0.83 (0.51 to 1.37)

HR, hazard ratio. The analysis that removed records of arrest after transplantation gave the same result as the main analysis because no event was based on such a record alone. Log-rank tests were as follows.

Factor	Men with factor, n	Events with factor	Men without factor, n	Events without factor	Log-rank p
Sensory profile	11	4	56	16	0.79
Autonomic profile	18	4	49	16	0.094
Motor, gait or balance profile	61	18	6	2	0.021
Adrenal insufficiency	36	13	31	7	0.37

Table S8 HPO subgroups at the index examination and later (n = 99 men with examination data)

HPO subgroup	Any record, n	Any record, %	At index examination, n	First recorded later, n
Cognitive	39	39.4	32	7
Psychiatric	28	28.3	17	11
Seizure	3	3.0	3	0
Cranial nerve	22	22.2	16	6
Movement or ataxia	54	54.5	46	8

Table S9 Most frequent HPO terms behind each cerebral subgroup (examination visits of men)

Subgroup	HPO term	Name	Examination visits with the term, n
Cognitive	HP:0011446	Abnormality of mental function	88
Cognitive	HP:0031987	Concentration or attention problems	77
Cognitive	HP:0001268	Mental deterioration	62
Psychiatric	HP:0000708	Atypical behavior	61
Psychiatric	HP:0000716	Depression	32
Psychiatric	HP:0007018	Attention deficit hyperactivity disorder	0
Seizure	HP:0001250	Seizure	14
Cranial nerve	HP:0001291	Abnormal cranial nerve morphology	14
Cranial nerve	HP:0002015	Dysphagia	13
Cranial nerve	HP:0000496	Abnormality of eye movement	12
Movement or ataxia	HP:0001317	Abnormal cerebellum morphology	123
Movement or ataxia	HP:0007010	Poor fine motor coordination	29
Movement or ataxia	HP:0001337	Tremor	12

Table S10 Landmark analysis of HPO findings at the index examination (n = 48, 12 events)

Finding at index examination	Men with finding, n	Events with finding	Events without finding	Hazard ratio (95% CI)	p	Adjusted p
Cognitive or psychiatric finding at first examination	5	2	10	4.40 (0.81 to 24.07)	0.087	0.24
Movement or ataxia finding at first examination	16	4	8	1.61 (0.43 to 6.01)	0.48	0.48
Cranial nerve finding at first examination	2	1	11	3.41 (0.41 to 28.36)	0.26	0.34
Any cerebral HPO finding at first examination	20	6	6	3.03 (0.74 to 12.38)	0.12	0.24